

Robotics & ROS WORKSHOP

A 5-Day Hands-On Robotics Training Program

Proposed by robotry | [Aryan Jagushte](#) | www.robotry.in

Workshop Objective

This 5-day workshop introduces students to Robotics and ROS2 (Robot Operating System) from the ground up. Concepts are introduced step by step and immediately applied through hands-on projects, so participants leave with practical, real-world robotics experience rather than just theory.

Workshop Flow

The workshop is structured to build knowledge progressively, from fundamentals to a working hardware demonstration:

- Basics of Robotics and ROS2
- Ubuntu Commands
- ROS2 Architecture
- Custom Packages and Nodes
- Robot Design
- Simulation of Robot
- Hardware Demonstration

Day-wise Breakdown

Day 1: Fundamentals of Robotics and ROS

- Introduction to Robotics & ROS2
- Why ROS2? Use-cases and Industry Trends
- Ubuntu Installation
- ROS2 Installation and Configuration
- Python Programming Essentials
- Basic Linux and ROS2 Commands

Day 2: ROS2 Core Concepts

- ROS2 Architecture: ROS2 Workspace and Package Structure
- Build your Workspace and create Package Creation using `ament_cmake`
- What are Nodes, Topics, Services, Actions
- Creating Custom Nodes: Hands-on practice with Publisher and Subscriber nodes.
- Launch Files
- What is DDS?, ROS_DOMAIN_ID

- ROS2 Simulation: Teleoperation and simulation using Turtlesim.
- Debugging Tools: Using `rqt_graph`, `ros2 topic info`, and other CLI commands to visualize the data flow.

Day 3: Robot Design and Simulation

- Design on Fusion 360: Modeling the basic robot
- URDF Conversion: Introduction to URDF and exporting the 3D model for simulation.
- Ubuntu Basics: Essential Linux commands for navigating the terminal.
- Simulation setup in gazebo, robot state publisher and spawning
- Rviz2 and TF2 view_frames (base_link)
- Gazebo Plugins and Odometry, diff_drive, lidar plugins
- Creating Custom World; GazeboSim working

Day 4: SLAM and Navigation

- Teleoperation: Driving their custom simulated robot using keyboard : teleop.
- Apply SLAM & Nav2 stack in the custom environment
- Localization and map_saver
- Navigation Parameters, costmap, behaviour tree, lifecycle nodes

Day 5: Bridging Hardware and ROS2

- ROS2 Communication with Microcontroller (Arduino Uno), serial communication
- Computer Vision: OpenCV and ROS2 Introduction
- Squarobot AMR Demonstration
- Q&A and Discussion

Workshop Deliverables

- Training with Real-Time Application
- Practical and Hands-On Exposure

Quotation

- The workshop excludes the cost of food and accommodation, and it should be provided by the college to the robotics team. For travelling expenses it should be provided by the college as per the ticket or else as mentioned below.

Training Mode	Duration	Fees (INR)	Travelling (INR)	Total (INR)
Offline	5 Days	₹40,000	₹5,000	₹45,000

About robotry

Robotry is an initiative by [Aryan Jagushte](#), a robotics engineer with a passion for making robotics education simple and impactful. Robotry has conducted numerous workshops and continues to bridge the gap between academia and real-world robotics.

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